

Advanced Statistical Methods II

Lecture 8: Time-to-event Data, Hazard, and Censoring

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Outline of Today's Lecture

Motivation

Time-to-event data

Survival function and related quantities

Censoring

Independent censoring

Likelihood contributions

Risk sets and first glimpse of Kaplan–Meier

Common pitfalls and interpretation

Practice questions

Key takeaway messages

In the first part of the course, we have repeatedly seen that the data we wish we had and the data we actually observe may be different.

- ▶ Missing covariate/response values led us to missing-data mechanisms.
- ▶ Latent labels led us to EM for mixture models.
- ▶ Incomplete cell counts led us to EM for tables.
- ▶ Difficult objective functions led us to MM algorithms.

Today we meet another major kind of incomplete observation:
time-to-event data with censoring.

Survival analysis is a systematic likelihood-based framework for such data ([Kleinbaum and Klein, 2012](#); [Kalbfleisch and Prentice, 2002](#)).

Suppose the response variable is a waiting time:

T = time from a well-defined origin to a well-defined event.

Examples:

- ▶ time from admission to discharge;
- ▶ time from treatment to death or recovery;
- ▶ time from installation to machine failure.

If some individuals have not yet experienced the event when the study ends, replacing their event time by the last follow-up time is wrong: the last follow-up time is not the true event time.

A simple motivating dataset



Consider a small clinical follow-up study.

Patient	Follow-up time	Status	Meaning
1	4	1	relapse observed at time 4
2	6	0	relapse-free until time 6
3	7	1	relapse observed at time 7
4	9	0	relapse-free until time 9
5	10	1	relapse observed at time 10

Here status is

$$\delta_i = \begin{cases} 1, & \text{event observed,} \\ 0, & \text{right censored.} \end{cases}$$

What should we do with patients 2 and 4?

What censored observations tell us



For patient 2, the recorded value is

$$(U_2, \delta_2) = (6, 0).$$

This does **not** mean

$$T_2 = 6.$$

It means

$$T_2 > 6.$$

Similarly, for patient 4,

$$(U_4, \delta_4) = (9, 0) \implies T_4 > 9.$$

A censored observation is incomplete but still informative. It gives an inequality rather than an exact value.

Today's main questions



1. What exactly is time-to-event data?
2. What are survival, hazard, and cumulative hazard functions?
3. How should we interpret the hazard function?
4. What are right censoring, left censoring, and interval censoring?
5. What does independent or noninformative censoring mean?
6. How do censored observations contribute to likelihoods?

Time-to-event variable?



A time-to-event variable is a nonnegative random variable

$$T \geq 0,$$

where T measures the time until a specified event occurs.

A survival-analysis problem must specify three ingredients clearly:

1. the time origin;
2. the event of interest;
3. the time scale.

Example

In a cancer study, the time origin may be date of surgery, the event may be recurrence, and the time scale may be months after surgery.

If every subject is followed until the event occurs, we observe

$$T_1, T_2, \dots, T_n.$$

Then standard distributional ideas apply:

- ▶ empirical distribution of event times;
- ▶ mean or median event time;
- ▶ parametric models such as exponential or Weibull;
- ▶ ordinary likelihood based on densities.

Difficulties in survival analysis: the response is positive, might be even skewed; and some event times are only partially observed.

Distribution function and survival function



Let T be a nonnegative event time.

The distribution function is

$$F(t) = \mathbb{P}(T \leq t), \quad t \geq 0.$$

The survival function is

$$S(t) = \mathbb{P}(T > t) = 1 - F(t).$$

$S(t)$ is the probability that an individual survives beyond time t , or equivalently, remains event-free beyond time t .

Typical properties:

- ▶ $S(0) \leq 1$, often $S(0) = 1$;
- ▶ $S(t)$ is nonincreasing in t ;
- ▶ $S(t) \rightarrow 0$ as $t \rightarrow \infty$.

Density and survival: two equivalent views



If T is continuous with density $f(t)$, then

$$F(t) = \int_0^t f(u) du.$$

Hence

$$S(t) = 1 - F(t) = \int_t^\infty f(u) du.$$

Also,

$$f(t) = F'(t) = -S'(t).$$

Interpretation

The density describes where event times tend to fall. The survival function describes how many individuals remain event-free beyond each time point.

The hazard function



The hazard function is defined as

$$h(t) = \lim_{\Delta t \downarrow 0} \frac{\mathbb{P}(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t}.$$

For continuous T , this becomes

$$h(t) = \frac{f(t)}{S(t)}.$$

The hazard is the instantaneous event rate at time t , among those who have survived up to time t (Cox and Oakes, 1984; Kleinbaum and Klein, 2012).

A hazard is not a probability. It is a rate and may exceed 1 depending on the time scale.

Deriving $h(t) = f(t)/S(t)$

Start with the conditional probability over a short interval:

$$\mathbb{P}(t \leq T < t + \Delta t \mid T \geq t) = \frac{\mathbb{P}(t \leq T < t + \Delta t)}{\mathbb{P}(T \geq t)}.$$

For small Δt ,

$$\mathbb{P}(t \leq T < t + \Delta t) \approx f(t)\Delta t.$$

Also, for continuous T ,

$$\mathbb{P}(T \geq t) = \mathbb{P}(T > t) = S(t).$$

Therefore,

$$\frac{\mathbb{P}(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t} \approx \frac{f(t)}{S(t)}.$$

Taking $\Delta t \downarrow 0$ gives

$$h(t) = f(t)/S(t).$$

How to read a hazard function



Suppose $h(12) = 0.04$ per month.

- ▶ This does **not** mean a probability of 0.04 over the whole study.
- ▶ It means that near month 12, among those still event-free, the instantaneous event rate is about 0.04 per month.
- ▶ Over a very short interval Δt ,

$$\mathbb{P}(12 \leq T < 12 + \Delta t \mid T \geq 12) \approx 0.04\Delta t.$$

The phrase “among those still at risk” is essential. Hazards are conditional on survival up to that time.

The cumulative hazard function is

$$H(t) = \int_0^t h(u) \, du.$$

Using $h(t) = f(t)/S(t) = -S'(t)/S(t)$,

$$h(t) = -\frac{d}{dt} \log S(t).$$

Integrating from 0 to t , assuming $S(0) = 1$,

$$H(t) = -\log S(t).$$

Equivalently,

$$S(t) = \exp\{-H(t)\}.$$

Survival vs hazard: different questions



Quantity	Question answered
$S(t) = \mathbb{P}(T > t)$	What proportion remains event-free beyond time t ?
$f(t)$	Around which times do events tend to occur?
$h(t)$	Among those still at risk at time t , how fast are events occurring now?
$H(t)$	How much event risk has accumulated up to time t ?

Example: exponential survival



If

$$T \sim \text{Exponential}(\lambda), \quad \lambda > 0,$$

then

$$f(t) = \lambda e^{-\lambda t}, \quad S(t) = e^{-\lambda t}.$$

Thus

$$h(t) = \frac{f(t)}{S(t)} = \frac{\lambda e^{-\lambda t}}{e^{-\lambda t}} = \lambda.$$

The exponential distribution has a constant hazard. It is useful as a starting model, but constant hazard is often unrealistic in medical and reliability applications.

Example: Weibull survival



One common parametrization of the Weibull distribution is

$$S(t) = \exp\{-\lambda t^\alpha\}, \quad \lambda > 0, \alpha > 0.$$

Then

$$H(t) = \lambda t^\alpha,$$

so

$$h(t) = H'(t) = \lambda \alpha t^{\alpha-1}.$$

Parameter value	Hazard shape
$\alpha = 1$	constant hazard
$\alpha > 1$	increasing hazard
$0 < \alpha < 1$	decreasing hazard

Reliability intuition

An increasing hazard may represent ageing or wear-out. A decreasing hazard may represent early failures among fragile units.

Quick check: survival, density, hazard



Suppose

$$S(t) = \exp(-2t^3), \quad t \geq 0.$$

Find $H(t)$, $h(t)$, and $f(t)$.

Solution.

$$H(t) = -\log S(t) = 2t^3.$$

Hence

$$h(t) = H'(t) = 6t^2.$$

Finally,

$$f(t) = h(t)S(t) = 6t^2 \exp(-2t^3).$$

What is censoring?



Censoring occurs when the event time T is not observed exactly, but partial information about T is available.

A censored observation provides a set of possible values for T , rather than one exact value.

Typical forms:

- ▶ right censoring: $T > C$;
- ▶ left censoring: $T < C$;
- ▶ interval censoring: $L < T \leq R$.

Censoring is not the same as missingness with no information. It is partial information about an event time.

Right censoring is the most common case in introductory survival analysis.

We observe

$$U = \min(T, C), \quad \delta = \mathbb{I}(T \leq C),$$

where

- ▶ T is the event time;
- ▶ C is the censoring time;
- ▶ $\delta = 1$ means the event was observed;
- ▶ $\delta = 0$ means the subject was censored.

If $\delta = 0$, then

$$U = C < T,$$

so we know

$$T > U.$$

Right-censored data are usually recorded as pairs (U_i, δ_i) , not simply as observed times.

Common sources of right censoring



Right censoring can arise in several ways.

Source	Interpretation
Administrative end	study ends before event occurs
Loss to follow-up	subject leaves study before event is observed
Dropout	subject stops participating
Competing practical end	device removed before failure
No event by analysis date	event may occur later, but not yet observed

The statistical validity of standard methods depends on how censoring occurs. Loss to follow-up caused by worsening health may be informative.

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- ▶ Infection occurred before the first positive test, but the exact infection time is unknown.
- ▶ A machine was already defective when inspected for the first time.

Left censoring gives an upper bound on T . Right censoring gives a lower bound on T .

Interval censoring

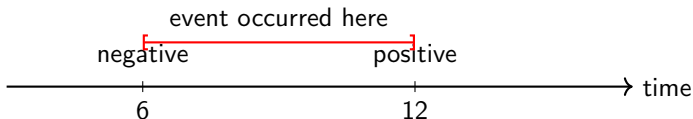


Interval censoring occurs when we know that the event time lies in an interval:

$$L < T \leq R.$$

Example: periodic medical screening

A patient tests negative at month 6 and positive at month 12. If the event is infection, then the infection time is known to lie between months 6 and 12.



Interval censoring is common whenever subjects are inspected periodically rather than continuously.

Censoring vs truncation



Censoring and truncation are different concepts.

Mechanism	What happens?
Censoring	subject is in the sample, but exact event time is not fully observed
Truncation	subject is not included in the sample unless their event time satisfies a condition

Example of left truncation

A study enrolls only people who have survived long enough to attend a clinic visit. Individuals who had the event before the clinic visit are absent from the dataset.

Today we focus mainly on censoring. Truncation needs different likelihood adjustments and should not be treated as ordinary censoring.

Classify each as right censored, left censored, interval censored, or truncated.

1. A patient is event-free at the final study visit at month 24.
2. A disease was already present at the first diagnostic test.
3. A subject tested disease-free in January and diseased in April.
4. Only individuals who survived at least 5 years after diagnosis are included in a retrospective dataset.
5. A machine is still working when the reliability experiment ends.

An exercise: answers



1. Right censored: the event time exceeds 24 months.
2. Left censored: the event time occurred before the first test.
3. Interval censored: the event occurred between January and April.
4. Left truncated: inclusion requires surviving at least 5 years.
5. Right censored: the failure time exceeds the end of the experiment.

A useful rule: censoring means partial information for a sampled unit; truncation affects whether the unit enters the sample at all.

Why censoring assumptions matter



Survival methods do not magically solve every incomplete follow-up problem.

The standard analysis of right-censored data usually assumes that censoring is independent or noninformative in an appropriate sense ([Kalbfleisch and Prentice, 2002](#); [Kleinbaum and Klein, 2012](#)).

Informally, independent censoring means that, after accounting for modeled covariates if any, the censoring mechanism does not carry additional information about the subject's event time.

If sicker patients are more likely to drop out, treating dropout as ordinary independent censoring may bias survival estimates.

A mathematical formulation



Let

- ▶ T be the event time;
- ▶ C be the censoring time;
- ▶ X be observed covariates.

A common condition is conditional independence:

$$T \perp C \mid X.$$

For simple one-sample survival analysis without covariates, this reduces conceptually to

$$T \perp C.$$

The censoring time may depend on observed covariates, but after conditioning on those covariates it should not depend on the unobserved event time in an unexplained way.

Independent vs informative censoring



More plausible independent censoring	Possible informative censoring
Study ends on a fixed administrative date	Patients drop out because symptoms worsen
Device is removed due to unrelated relocation	Machine is removed because it is visibly deteriorating
Follow-up stops due to funding end date	High-risk participants become harder to contact

Relation to missing-data mechanisms



There is a strong conceptual link with Lectures I–II.

Missing data language	Survival language
Observed value missing	Exact event time not observed
Missingness mechanism	Censoring mechanism
Ignorable missingness under conditions	Noninformative/independent censoring under conditions
Complete-case deletion may bias estimates	Deleting censored observations may bias survival estimates

Censoring is not arbitrary missingness. It has a special structure: the censored record usually tells us that T_i is above or below some value.

Why likelihoods handle censoring naturally



Likelihoods are based on the probability of the data we actually observed.

For an exact observed failure at time t , we observe

$$T = t.$$

So the likelihood contribution is based on

$$f(t; \theta).$$

For a right-censored observation at time c , we observe

$$T > c.$$

So the likelihood contribution is based on

$$S(c; \theta) = \mathbb{P}_\theta(T > c).$$

The likelihood contribution must match the information contained in the observation: exact time gives a density; right censoring gives a survival probability.

Right-censored likelihood: preview



For independent observations (U_i, δ_i) ,

$$U_i = \min(T_i, C_i), \quad \delta_i = \mathbb{I}(T_i \leq C_i),$$

the event-time part of the likelihood is

$$L(\theta) = \prod_{i=1}^n f(U_i; \theta)^{\delta_i} S(U_i; \theta)^{1-\delta_i}.$$

Equivalently, since $f(t) = h(t)S(t)$,

$$L(\theta) = \prod_{i=1}^n \{h(U_i; \theta)S(U_i; \theta)\}^{\delta_i} S(U_i; \theta)^{1-\delta_i}.$$

Therefore,

$$L(\theta) = \prod_{i=1}^n h(U_i; \theta)^{\delta_i} S(U_i; \theta).$$

Likelihood contributions by censoring type



Observation type	Information observed	Contribution
Exact event	$T = t$	$f(t)$
Right censored	$T > c$	$S(c)$
Left censored	$T \leq c$	$F(c)$
Interval censored	$L < T \leq R$	$F(R) - F(L)$

This table is one of the main bridges from today's lecture to the next lecture on censored likelihoods.

Worked mini-example: write the likelihood



Suppose $T \sim \text{Exponential}(\lambda)$ and we observe

$$(4, 1), \quad (6, 0), \quad (7, 1), \quad (9, 0), \quad (10, 1).$$

For the exponential distribution,

$$f(t; \lambda) = \lambda e^{-\lambda t}, \quad S(t; \lambda) = e^{-\lambda t}.$$

The likelihood is

$$L(\lambda) = \{\lambda e^{-4\lambda}\} \{e^{-6\lambda}\} \{\lambda e^{-7\lambda}\} \{e^{-9\lambda}\} \{\lambda e^{-10\lambda}\}.$$

Thus

$$L(\lambda) = \lambda^3 \exp\{-\lambda(4 + 6 + 7 + 9 + 10)\}.$$

What would go wrong if we ignored censoring?



For the same data:

$(4, 1), (6, 0), (7, 1), (9, 0), (10, 1),$

there are two common mistakes.

1. Treat 6 and 9 as event times.
2. Delete the censored observations $(6, 0)$ and $(9, 0)$.

Both strategies distort the information. Censored observations are neither exact failures nor useless records.

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For exponential data, the correct likelihood uses

number of observed events = 3, total observed follow-up time = 36.

So the MLE is

$$\hat{\lambda} = \frac{3}{36} = 0.0833.$$

For right-censored exponential data,

$$L(\lambda) = \prod_{i=1}^n \{\lambda e^{-\lambda U_i}\}^{\delta_i} \{e^{-\lambda U_i}\}^{1-\delta_i}.$$

Combine the powers of λ and the exponential terms:

$$L(\lambda) = \lambda^{\sum_i \delta_i} \exp \left\{ -\lambda \sum_{i=1}^n U_i \right\}.$$

Therefore the log-likelihood is

$$\ell(\lambda) = \left(\sum_i \delta_i \right) \log \lambda - \lambda \sum_i U_i.$$

Deriving the exponential MLE with censoring: II



From the previous slide,

$$\ell(\lambda) = \left(\sum_i \delta_i \right) \log \lambda - \lambda \sum_i U_i.$$

Differentiating,

$$\frac{d\ell}{d\lambda} = \frac{\sum_i \delta_i}{\lambda} - \sum_i U_i.$$

Setting this equal to zero gives

$$\frac{\sum_i \delta_i}{\lambda} = \sum_i U_i.$$

Thus

$$\hat{\lambda} = \frac{\sum_i \delta_i}{\sum_i U_i}.$$

Deriving the exponential MLE with censoring: III



To check that the critical point is a maximum, differentiate once more:

$$\frac{d^2\ell}{d\lambda^2} = -\frac{\sum_i \delta_i}{\lambda^2}.$$

If at least one event is observed, then $\sum_i \delta_i > 0$, so

$$\frac{d^2\ell}{d\lambda^2} < 0.$$

Therefore the critical point is a maximum.

For exponential right-censored data,

$$\hat{\lambda} = \frac{\text{number of observed events}}{\text{total observed follow-up time}}.$$

We write

At an observed event time t_j , define

- Risk sets are the bookkeeping device that lets us compare events with the subjects who were actually available to experience those events.

Risk sets for the motivating dataset



For the data

$(4, 1), (6, 0), (7, 1), (9, 0), (10, 1),$

observed event times are 4, 7, 10.

Event time t_j	At risk n_j	Events d_j	Conditional event fraction
4	5	1	1/5
7	3	1	1/3
10	1	1	1/1

Why is $n_j = 3$ at time 7?

Patient 1 already had the event at 4. Patient 2 was censored at 6. Patients 3, 4, and 5 are still at risk just before 7.

A preview of the product-limit idea



At event time t_j , the estimated conditional probability of surviving past that event time is

$$1 - \frac{d_j}{n_j}.$$

Multiplying these conditional survival probabilities gives the Kaplan–Meier estimator:

$$\hat{S}(t) = \prod_{t_j \leq t} \left(1 - \frac{d_j}{n_j}\right).$$

This is only a preview. We will derive and study the Kaplan–Meier estimator carefully in next week.

A Question



A subject has status $\delta = 0$ at time 8. Another subject has status $\delta = 1$ at time 8.

Do the two records contain the same information? Are their likelihood contributions same?

A subject has status $\delta = 0$ at time 8. Another subject has status $\delta = 1$ at time 8.

Do the two records contain the same information? Are their likelihood contributions same?

Solution. No.

- ▶ $(8, 1)$ means an event occurred exactly at time 8.
- ▶ $(8, 0)$ means the subject was event-free through time 8 and the event time is greater than 8.

Their likelihood contributions are different:

$$(8, 1) : f(8), \quad (8, 0) : S(8).$$

The same recorded time can mean different things depending on the event indicator.

Key takeaway messages



1. Time-to-event data require a clear origin, event, and time scale.
2. The survival function is $S(t) = \mathbb{P}(T > t)$.
3. The hazard $h(t) = f(t)/S(t)$ is an instantaneous event rate among those still at risk.
4. Censoring gives partial information about T ; it is not the same as observing T exactly.
5. Right-censored data are commonly recorded as (U_i, δ_i) .
6. Independent/noninformative censoring is a key assumption behind standard survival methods.
7. Likelihood contributions must reflect what was actually observed.

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